2025 IEEE Integrated STEM Education Conference (ISEC) | 979-8-3315-1343-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/ISEC64801.2025.11147331

STEM Education in the Age of Generative AI: Case Studies in Teaching Mathematics and Computer Science to Generation Z and Alpha

1st Myo Thida
Dept. of Computer Science
Bard College at Simon's Rock
Great Barrington, USA
ORCID: 0000-0002-5259-4010

2nd Amanda Landi
Dept. of Mathematics
Bard College at Simon's Rock
Great Barrington, USA
ORCID: 0000-0003-1847-4450

***Abstract*—This paper presents two pedagogical approaches for teaching in the era of generative artificial intelligence (GenAI), using case studies from mathematics and computer science. As Generation Z and Alpha have grown up immersed in technology, their learning styles differ significantly from previous generations, creating challenges for traditional instructional methods. While digital platforms provide flexibility, accessibility, and interactivity, the rapid advancement of generative AI adds new layers of complexity. These AI tools enhance self-learning and problem-solving but also raise concerns about ethical usage and conceptual depth. Building on existing research, we introduce two methods: one based on a "learning how to learn" approach in mathematics, which emphasizes mathematical reasoning and problem interpretation over procedural mastery, and another for integrating AI into computer science to support problem-solving, software development, and data-driven decision-making. Preliminary results suggest that students using these approaches demonstrate a stronger ability to leverage AI responsibly, critically assess its outputs, and apply it across disciplines. These findings highlight the need for curriculum adaptations that foster critical thinking, ethical awareness, and adaptability in the AI-driven future.**

***Index Terms*—STEM Education, Generative AI, Gen Z and Alpha, Ethical AI**

I. Introduction

Early research suggests that the learning styles of Generation Z and Alpha differ significantly from those of earlier generations, largely due to their familiarity with technology [1]. Consequently, maintaining student engagement in the classroom has become increasingly difficult.

Traditional in-class lectures lack the flexibility of online resources; for instance, students cannot rewind a live lecture if they lose focus momentarily, whereas platforms such as Khan Academy allow them to do so. Similarly, in-person lectures cannot be slowed down to accommodate individual learning needs, whereas educational content creators, such as the Organic Chemistry Tutor, provide such options. Moreover, online learning materials are often more accessible for students with diverse learning needs, offering features such as closed captioning and adjustable playback speeds. Additionally, digital content is frequently enhanced with engaging elements, such as vibrant visuals and auditory effects, which contribute to a more stimulating learning experience.

With continuous advancements, generative AI has taken technological evolution to the next level, introducing new challenges and opportunities in education. For instance, AI-powered tools like Symbolab [2] can now explain fundamental mathematical concepts step by step, making it easier for students to grasp complex topics. Similarly, GitHub Copilot [3] can generate codes with greater efficiency than a junior programmer, complete with clear comments and documentation. While these tools provide invaluable support for self-learning, they also raise critical challenges regarding ethical use and the development of a deep conceptual understanding.

These challenges place growing responsibilities on educators, especially in the early years of college when students are building foundational knowledge. While most educators in academia today belong to Generation Y (Millennials), X, or even the Baby Boomer generation, the majority of students are from Generation Z and the emerging Generation Alpha—groups that have grown up with technology as an integral part of their daily lives.

This generation difference influences learning expectations and engagement with AI-driven tools. Since AI can easily handle tasks at the 'remembering' level of Bloom's Taxonomy, traditional rote learning, such as repetitive math exercises or basic programming practice, is becoming less relevant and less appealing to students. Instead, the focus must shift toward helping students engage in deeper levels of learning—understanding, applying, analyzing, and creating. Educators must design curriculum and assessments that encourage students to think critically, evaluate AI-generated solutions, and integrate AI as a tool for exploration rather than a shortcut for answers.

These shifts in learning dynamics are particularly evident in STEM education, where AI tools are transforming both how students approach learning and how educators design their instruction. To explore this transformation, this paper examines case studies in mathematics and computer science. While mathematics education has historically emphasized conceptual understanding and problem-solving, the availability of AI tools reshapes how students engage with learning, shifting the focus from procedural mastery to mathematical reasoning, problem interpretation, and comprehension of word problems. Similarly, computer science education is no longer just about programming and algorithmic thinking; it now involves teaching students how to integrate AI into problem-solving, software development, and data-driven decision-making.

By analyzing these case studies, this paper aims to illustrate how generative AI is shaping the educational landscape and propose strategies for adapting curricula and teaching methodologies. The goal is not to replace traditional instruction but to teach students how to gain a deeper understanding by leveraging AI and existing knowledge, while preparing them to navigate an AI-driven future with critical thinking and adaptability.

Preliminary results from these case studies indicate that students exposed to a metacognitive approach in mathematics exhibit stronger problem-solving skills, improved reasoning, and enhanced communication of complex ideas. Those engaged in AI-integrated coursework demonstrate a more critical approach to AI-generated insights, actively assessing reliability, adapting models to real-world scenarios, and incorporating ethical considerations into their analyses. Early observations from interdisciplinary projects, such as a recent datathon competition, suggest that students who have experienced both pedagogical approaches are better equipped to integrate AI responsibly while leveraging its strengths for problem-solving across domains.

The remainder of this paper is organized as follows. Section II reviews relevant literature, focusing on state-of-the-art research on AI in education. Section III presents two case studies in STEM education, analyzing their implementation, impact, challenges, and outcomes. Finally, Section V summarizes key findings, discusses limitations, and outlines directions for future research.

II. Literature Review

The ”Artificial Intelligence in Education: A Review” [4] by L. Chen, P. Chen, and Z. Lin serves as a foundational resource for understanding the current landscape of AI in educational settings. The authors provided a broad overview of AI's role in education, exploring various applications such as intelligent tutoring systems, adaptive learning platforms, and automated grading. Their review highlights both the potential benefits and challenges of integrating AI in educational environments, including the enhancement of personalized learning and efficiency in administrative tasks. However, the authors did not address the potential ethical concerns and limitations associated with these technologies.

In contrast, Holmes and Tuomi [5] provided a comprehensive review of existing AI tools in education (AIED) as of 2022, highlighting that many AI applications are based on a limited view of the functions of education—such as focusing solely on measurable outcomes. They also emphasize the ethical concerns related to using AI in education, including how AI technologies are primarily trained on American English and the potential issues of imposing these tools on the Global South (developing countries).

However, both review papers acknowledged the adoption of AI technologies in education while emphasizing the need for caution. Educators have historically approached educational technology and AI-driven tools with caution. Yet, the emergence of Generative AI models, such as ChatGPT, in 2022 led to a sharp rise in both interest and concern regarding AI's role in education. Many educators and the public became increasingly aware of generative AI's ability to write essays, complete assignments, and generate responses with just a well-crafted prompt. At the same time, concerns over AI hallucinations—where AI generates misleading or incorrect information while presenting it as factual—became even more pressing.

In response to the rise of Generative AI, educators have developed a variety of strategies. Some decided to ban the use of Generative AI in the classroom, while others allowed students to use it, provided they declared its use in their assignments. Pedagogies and lesson plans that integrate AI usage are actively being discussed. In 2023, the U.S. Department of Education published a report on "Artificial Intelligence and the Future of Teaching and Learning" [6], offering seven actionable recommendations for education leaders to guide AI's integration into the educational landscape. Complementing this, UNESCO's AI competency frameworks for students and teachers emphasize critical skills in human-centered thinking, ethical considerations, foundational AI applications, and system design [7]. Both policy papers indicate that Generative AI and broader AI technologies are now integral to education, and educators must develop strategies to leverage the resources generated by AI while teaching students to remain critical of AI hallucinations.

Despite these policy efforts, there remains a limited body of literature specifically addressing the effects of Generative AI and documenting the strategies educators are employing to address the ethical challenges of its use. Given the limited literature on this topic, it is crucial to document and analyze the various approaches being adopted by educators to better understand the challenges and best practices for integrating Generative AI into education.

III. Case Studies

To address the evolving challenges in student learning, we explore two implementation strategies in STEM education. The first focuses on teaching students how to learn mathematics by integrating writing as a tool for metacognitive development. The second presents an approach to incorporating generative AI into curriculum design, equipping students with the skills to critically engage with AI-driven tools. Both strategies aim to help students leverage existing technologies effectively while fostering deep content understanding, ethical integration, and critical thinking.

A. Case Study I: Learning How to Learn in Math Class

We believe the solution to these challenges is to move away from teaching students math content and move towards teaching students *how to learn* mathematics. With this focus, students will be able to learn mathematics no matter the format the content is delivered. Moreover, these skills are transferrable to any field of study and will help students adapt in the future when new technologies will inevitably force them to confront their current knowledge and skill set. We have tried at least two designs to emphasize *how to learn* mathematics. We say more below about our implementation of writing as a tool to train a student's metacognitive skills.

Writing in math class is well-documented as a way to help students cultivate their metacognition and to be more aware of themselves as learners, e.g. [8]. To determine its effectiveness at the undergraduate level, we designed curricula for a precalculus and a Calculus I course in which writing is the primary mode of engagement with the mathematics [9], [10]. While we did have to sacrifice some content in order to focus on writing in the classroom, we found that several students in each course improved their ability to think about and to write about mathematics. Qualitative evidence is available in both [9], [10].

Several of the writing activities were given in the classroom as well as for homework, but only a subset of those need to be collected for feedback by the instructor. The writing activities are designed in such a way that students are less likely to revert to online technologies, such as videos, intelligent tutors, or calculators. That is, they tend to be more open-ended or to be more personal. Interestingly, students preferred to participate in the writing in the classroom to be turned in (in the subset of written responses collected, as described above) because that reduced their workload outside of class. As a result, more students were engaged in these activities.

In this online blog post, author Amanda Landi shares a specific example of an in-class script (see Figure 2 in the source). Observe that the earlier part of the script, giving relevancy to the prior night's homework assignment, which had students focus on their understanding of a relationship between functions, then has students attempt math but has them concentrate on their own learning process, their mistakes. After students focus on their own higher-order thoughts, the instructor then goes over the lower-level math alongside students so they still practice the basic portion of the lesson.

Lessons such as this helped to engage students. There are several shorter activities, requiring students to think critically and to respond. Some prompts are more open-ended, allowing students who do not come prepared to still participate in the learning inside the classroom.

While this method is not without its own challenges, it does train students metacognitive skills, putting them on the pathway of becoming more autonomous learners. In fact, this kind of experiential-based learning is recommended for generation Alpha [1].

B. Case Study II: Incorporating Generation AI in the Curriculum and Lesson Plan

Computer Science is one of the fields where industrial experience (vocational training) can sometimes surpass academic qualifications. With Generative AI capable of producing industry-grade code, traditional methods of teaching basic concepts and theory may become obsolete. According to the AI Index report by Stanford, in 2024 [11], the industry produced 51 notable machine learning models, while academia contributed only 15. This shift underscores the need for academia to emphasize application-driven learning and conceptual understanding, even at the early college level.

Recognizing this, institutions worldwide are rethinking how AI should be integrated into education. In September 2024, UNESCO launched two AI competency frameworks—one for students and another for teachers—aiming to equip students with essential AI-related skills that go beyond memorization. The AI competency framework for students emphasizes four core competencies:

- **A human-centered mindset**: Encouraging students to understand and assert their role in relation to AI.
- **Ethics of AI**: Teaching responsible use, ethics-by-design, and safe practices.
- **AI techniques and applications**: Providing foundational knowledge and skills related to AI.
- **AI system design**: Fostering problem-solving, creativity, and design thinking.

In response to these shifts, we adapted the Computer Science curriculum to emphasize the application of AI tools in a responsible and ethically aware manner. For example, in the Machine Learning Class, we implemented a 90-minute lesson plan (Table I) that guides students through the integration of AI-driven code generation tools like GitHub Copilot and ChatGPT. This lesson plan not only teaches students how to use these tools effectively but also encourages them to critically evaluate the generated code, ensuring that they approach AI solutions with a human-centered mindset.

This lesson plan serves as an early foundation in the course, progressively building the skills necessary for students to independently design and develop their own AI tools tailored to their chosen fields by the end of the semester. This curriculum equips students with the following:

Human-Centered Mindset: By encouraging students to view AI-generated solutions as tools requiring human oversight and adaptation, the curriculum instills the importance of integrating human judgment into AI workfellow. This emphasizes that **AI should complement, not replace, human decision-making**, ensuring solutions align with real-world contexts and human values.

Ethics of AI: Identifying and documenting issues encountered with AI solutions teaches students accountability and awareness of potential biases or flaws in AI systems. This practice nurtures a sense of responsibility in designing and deploying ethical AI applications, preparing students to

TABLE I
90-MINUTES LESSON PLAN

Time	Content
15 minutes	Review of Regression Problems Each student briefly presents their regression problem.
20 minutes	Introduction to AI-Generated Code Lecture: Overview of how AI tools (co-pilot, GPT 4.0) can generate code for regression tasks. Activity: Develop ML model using GTP 3.5 or Co-Pilot. Use following Prompt. Prompt: "Develop ML model for the regression."
20 minutes	Running on the local machine / Google Cloud Activity: Copy the code, run in their local machine or google cloud. Identify 3 problems they encountered. (If there is no problem, write down the prior installed libraries).
5 minutes	break
15 minutes	Discussion and Troubleshooting Activity: Work in Pair and discuss if they can run the codes in the local machine. Share experiences and solutions to common configuration issues in working on the local machine.
10 minutes	Group Discussion Activity: Discuss potential challenges to be faced while tuning AI-generated code to address the regression problem in hand. **Homework**: Tune the code based on their regression problems. Write down the process.

prioritize fairness, transparency, and societal impact in their work.

AI Techniques and Applications: The hands-on troubleshooting and adaptation of AI solutions enhance students' technical proficiency in understanding and applying AI techniques. By learning to tailor AI-generated code to specific environments, students gain **practical insights** into how AI applications operate, their limitations, and strategies for optimization.

C. Impacts and Outcomes

Students who had engaged with the "learning how to learn" mathematics approach exhibited stronger problem-solving skills, particularly in structuring their thought processes, documenting their reasoning, and effectively communicating complex ideas. Their metacognitive training enabled them to approach data analysis with a deeper understanding of patterns, assumptions, and logical frameworks, rather than relying solely on computational tools.

Similarly, students who had participated in the AI-integrated curriculum demonstrated a more critical approach to AI-generated insights. Rather than accepting outputs at face value, they actively assessed the reliability of AI-driven analyses, adjusted models for real-world applicability, and discussed ethical implications related to bias and transparency. Their ability to adapt AI tools to diverse dataset (vehicles classification, music genre prediction, gender classification, news article classification) and problem contexts highlighted their deeper technical understanding and ethical awareness.

These competencies were particularly evident in the recent datathon competition, where students collaborated across disciplines to analyze the election dataset, generate insights, and present their findings. Participants who had engaged with both pedagogical approaches exhibited a strong ability to critically assess outputs, integrate ethical considerations into their analyses, and communicate their reasoning effectively.

While comprehensive data collection to evaluate the impact of the proposed implementations is still in progress, preliminary observations indicate promising results. Students who completed both courses showed improved capacity to apply AI responsibly and leverage its strengths for problem-solving across disciplines.

These findings also highlight the importance of balanced pedagogy: ensuring that AI is used as an augmentation tool rather than a substitute for foundational learning. The *learning how to learn* approach discussed in Case Study 1 plays a crucial role in sharpening foundational learning, reinforcing students' ability to engage deeply with AI-driven methodologies without compromising their problem-solving skills. Further structured analysis will help solidify these findings and inform future curriculum enhancements.

IV. Future Work and Scalability

While the proposed methods have shown promise in small-class settings, scaling this approach to larger classes presents significant challenges. In large classrooms, fostering meaningful in-class discussions and ensuring individualized engagement can be difficult. Even within small institutions, scaling AI-integrated learning across multiple courses presents challenges that need to be carefully addressed.

Firstly, many educators and administrators may be hesitant to adopt AI-driven learning due to concerns about over-reliance on technology, ethical risks, or the disruption of traditional teaching methods. Gaining institutional support will require clear communication of the benefits, pilot programs to demonstrate effectiveness, and structured faculty training to ensure responsible implementation.

Secondly, since AI-integrated learning demands a shift in teaching methodology, faculty members need targeted training not just in AI tools but also in pedagogical strategies that balance AI use with foundational learning. Providing professional development workshops, peer mentoring programs, and access to shared teaching resources will be critical in ensuring a smooth transition.

Thirdly, expanding this approach institution-wide requires a well-structured curriculum that integrates AI thoughtfully across different disciplines while maintaining rigorous assessment standards. Developing adaptable modules that cater to various levels of AI literacy and ensuring robust evaluation methods will be essential to maintaining academic integrity.

Beyond these considerations, fostering collaboration between departments—such as Computer Science, Mathematics, and Humanities—can broaden the impact of AI-integrated learning. Additionally, incorporating real-world projects, such as data-driven policy analysis or ethical AI case studies, will help students develop practical skills and reinforce the responsible use of AI.

Although full-scale implementation remains an ongoing process, continued research and structured analysis of small-scale efforts will provide valuable information. Future work will focus on quantitative analysis of student performance metrics, engagement levels, and longitudinal studies to measure the effectiveness of AI-integrated learning. These findings will guide iterative improvements and inform best practices for broader adoption.

V. Conclusion

As educators, we have been teaching Generation Z for several years, and Generation Alpha is now entering higher education. While artificial intelligence has been present in academic settings for over a decade, the emergence of generative AI introduces both new challenges and opportunities. AI will not replace faculty, but it necessitates an adaptation of pedagogical approaches to align with the evolving learning styles of digital-native students. Rather than solely leveraging AI to reduce instructional workload—though it can assist with tasks such as auto-grading and administrative processes—the primary focus should be on equipping students with the skills to critically assess, interpret, and ethically engage with AI-generated content. By fostering these competencies, educators can ensure that students develop a deeper conceptual understanding and the ability to navigate an AI-driven academic and professional landscape with integrity and adaptability.

References

[1] Ziatdinov, R., & Cilliers, J. (2022). Generation Alpha: Understanding the next cohort of university students. arXiv preprint arXiv:2202.01422.

[2] Michal Avny, Adam Arnon, and Lev Alyshayev. Symobla, https://www.symbolab.com/.

[3] OpenAI and Microsoft, GitHub Co-Pilot, 2021 https://github.com/features/copilot

[4] Lija Chen, Pingping. Chen and Zhijian. Lin, Artificial Intelligence in Education: A Review, in IEEE Access, vol. 8, pp. 75264-75278, 2020, doi: 10.1109/ACCESS.2020.2988510

[5] Holmes, W., & Tuomi, I. (2022). AI and education: The importance of a socio-technical and critical approach. *European Journal of Education, 57*(4), 542-557. Wiley. https://doi.org/10.1111/ejed.12533

[6] Miguel A. Cardona, Ed.D., & Roberto J. Rodríguez, U.S. Department of Education, "Artificial intelligence and the future of teaching and learning," May 2023.

[7] F. Miao and K. Shiohira, AI Competency Framework for Students, 2024, corporate author: UNESCO [8657]. [Online]. Available: https://doi.org/10.54675/JKJB9835

[8] Hoffer, W. W. (2016). Developing literate mathematicians: A guide for integrating language and literacy instruction into secondary mathematics. National Council of Teachers of Mathematics, Incorporated.

[9] Landi, A., & Minden, K. (2024). Writing as Active Learning in Gateway Undergraduate Mathematics. Early College Folio, 3(1), 4.

[10] Landi, A. A Case Study of Writing as Active Learning in Gateway Undergraduate Mathematics: Model in Calculus I. The IWT CLASP Journal: Studies in Student-Centered Teaching and Learning around the Globe, Vol 1, (2024).

[11] Stanford University, AI Index Report, 2024, https://aiindex.stanford.edu/

Myo Thida , a Senior IEEE member, received the B.Eng. and M.Eng. degrees in electrical and electronic engineering from Nanyang Technological University, Singapore, in 2001 and 2005, respectively, and a Ph.D. in Computer Vision from Kingston University, London, UK, in 2013. Currently serving as an assistant professor of Computer Science at Bard College at Simon's Rock in the United States, she has previously contributed her expertise to the Computer Engineering Department at Chiang Mai University. From 2008 to 2013, she worked as a research engineer and scientist at A*STAR. Myo has published three technical books, over ten journal and conference papers, with a research h-index of 9 and an i-10 index of 9. Her current research interests include data science machine learning algorithms, and deep learning architectures. She is passionate about harnessing AI for real-life applications and creating impacts that benefit society. In addition to her technical work, she is actively involved in promoting STEM education in developing countries and advocating for data-driven policies.

Amanda Landi earned her PhD in Applied Mathematics from North Carolina State University. Currently, Amanda is an Associate Professor of Mathematics with Bard College at Simon's Rock, and she has been with Simon's Rock since 2015. During her tenure there, Amanda has written and published papers in the field of math education, trained educators in metacognitive and practice executive functioning skills building, and she has designed curriculum for math courses ranging from high school to upper level undergraduate mathematics. Other research projects are at the intersection of mathematics, machine learning, and data science.